Spherical Cyclic Time v1.5-F
Toy Kernel Refinement Closure Addendum

Spherical Cyclic Time / Tempo Sferico Ciclico project

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Abstract

This addendum closes the v1.5 perturbative-kernel refinement route for the Spherical Cyclic Time (SCT) toy/informational framework. The main update is narrow: the previously free soft-kernel exponent in $T_R(k) = \exp[-(k/k_c)^p]$ is restricted to the Gaussian/diffusive toy value $p = 2$. This improves the internal status of the transfer-kernel shape, but does not derive k_c, $P_\xi(k)$, or a physical cosmological restart. The work remains non-peer-reviewed and toy/informational only.

1 Purpose

The v1.5 branch asked whether the perturbative matching sector could be made less arbitrary without tuning the classifier. The old mainline used

$$T_R(k) = \exp[-(k/k_c)^p]. \tag{1}$$

The v1.5-B feasibility audit showed that an auxiliary diffusion or Gaussian coarse-graining transition gives

$$\partial_s \delta = D\nabla^2 \delta, \qquad T_R(k) = \exp[-D\Delta s\, k^2] = \exp[-(k/k_c)^2]. \tag{2}$$

Thus $p = 2$ is retained as a toy-derived kernel shape. General p is demoted to stress and sensitivity tests.

2 Restricted audit result

The v1.5-D restricted audit executed the preregistered $p = 2$ grid with no reopening of p, no selection of k_c, and no parameter tuning. The result was:

Classification	Count	Fraction
coherent_toy	31	0.0554
weak	269	0.4804
critical	136	0.2429
failed	124	0.2214

The coherent-toy region is therefore non-empty but small: $31/560 = 5.54\%$.

3 k_c closure

The v1.5-E scale audit checked candidate sources for k_c: diffusion length, Gaussian smoothing radius, horizon crossing, correlation length, equality/BAO scale, Silk damping, nonlinear/data-fitted scales, and ekpyrotic/bounce-inspired transfer. None can be promoted inside the present SCT framework without an independent dynamical model. Therefore k_c remains a full-grid or requirements-only composite scale.

4 Claim boundary

Allowed claims:

- $p = 2$ has a minimal toy derivation from Gaussian/diffusive smoothing.
- The restricted $p = 2$ grid has a reproducible non-empty coherent-toy region.
- The result improves only the toy status of the transfer-kernel shape.

Forbidden claims:

- no physical cosmological restart is derived;
- no microphysical k_c is derived;
- no $P_\xi(k)$ is derived;
- no observational prediction or evidence is claimed.

5 Publication update

If this addendum is included before deposition, the recommended public version is v1.5.0. The publication set should include the v1.4-H patched English and Italian papers, the v1.4-F reproducibility supplement, this v1.5-F addendum, and the v1.5-F source/closure package.

6 Closure statement

The v1.5 branch is closed conservatively. SCT now has a better-motivated toy transfer-kernel shape, but not a physical transfer function. Reopening the physical route requires an independent action/background/transition model that fixes k_c and $P_\xi(k)$ before classifier counts are inspected.